F12 4/n CAR

Lecture 22: Section 3.3
Properties of Logarithms

Properties:

$$\log_a(uv) = \log_a u + \log_a v$$

$$\log_a\left(\frac{u}{v}\right) = \log_a u - \log_a v$$

$$\log_a u^n = n \log_a u$$

Change of base formula

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Properties of Logarithms

Recall the following properties of Logarithm:

$x > 0$

The logarithmic function with base a $a > 0$, $a \neq 1$

$y = f(x) = \log_a x$ if and only if $a^y = x$.

1. Domain of f: $(0, \infty)$
2. $\log_a 1 = 0$
3. $\log_a a = 1$
4. $\log_a a^x = x$ for all real number x

   $a^{\log_a x} = x$ for $x > 0$

The Natural Logarithmic Function — base is e

$y = \ln x$ if and only if $e^y = x$ $\quad 2 < e < 3$

Note the following:

$\ln 1 = 0$ $\quad \ln e = 1$

$e^{\ln x} = x,\ x > 0$ $\quad \ln(e^x) = x$, for all x.

$y = \ln x = \log_e x$
$= \ln_e x$

Properties of Logarithms

Let u, v and a be positive real numbers with $a \neq 1$ and n be any real number. The following properties hold:

* memorize

1. $\log_a(uv) = \log_a u + \log_a v$
2. $\log_a\left(\frac{u}{v}\right) = \log_a u - \log_a v$
3. $\log_a u^n = n \log_a u$

Proof: $\log_a u^n = \log_a(\underbrace{u \cdot u \cdot \ldots u}_{n \text{ factors of } u})$

$\log_a u^n = \log_a u + \log_a u + \ldots + \log_a u$ n terms of the log

$\log_a u^n = n \log_a u$

NOTE: $\log_a(u+v) \neq \log_a u + \log_a v$ $\quad u+v \neq uv$

$(\log_a u)^n \neq n \log_a u$

$(\log_a u)^n \neq \log_a u^n = \log_a(u^n)$

$\log_a u \cdot \log_a u \cdot \ldots \cdot \log_a u$ n factors of log

ex. Evaluate:

$3 =$ 1) $\log_4 2 + \log_4 32 = \log_4(2 \cdot 32) = \log_4 64 = \boxed{3}$

2) $\log_2 80 - \log_2 5 = \log_2 \frac{80}{5} = \log_2 16 = \boxed{4}$

$-\frac{1}{2} =$ 3) $-\frac{1}{3}\log_4 8 = \log_4 8^{-1/3} = \log_4 \frac{1}{8^{1/3}} = \log_4 \frac{1}{\sqrt[3]{8}} =$

$= \log_4 \frac{1}{2} = \log_4 2^{-1} = y \quad 4^y = 2^{-1} \quad (2^2)^y = 2^{-1}$

$2^{2y} = 2^{-1}$

$2y = -1$

$y = \frac{-1}{2}$

ex. Rewrite and simplify if possible:

1) $\ln(2 + e^x)$ already simplified

2) $\log_2(x - y)$

3) $\dfrac{\log_3 x}{\log_3 y}, \ y \neq 1$

4) $\ln\left(\frac{1}{\sqrt[3]{e}}\right) = \ln 1 - \ln \sqrt[3]{e} = 0 - \ln e^{1/3}$

$= 0 - \frac{1}{3}\ln e$ $= 1$

$= 0 - \frac{1}{3} = \boxed{\frac{-1}{3}}$

5) $\log_9\left(\frac{\sqrt[4]{9}}{3}\right) = \log_9 \sqrt[4]{9} - \log_9 3 =$ $= \frac{1}{2}$ $= 1$

$= \log_9 9^{1/4} - \frac{1}{2} = \frac{1}{4}\log_9 9 - \frac{1}{2}$

$\frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$

6) $2^{4\log_2 x} = \cancel{2^{\log_2}} x^4$

$a^{\log_a x} = x$

$= x^4$

7) $\ln\sqrt{\frac{x^3y}{z}} = \ln\left(\frac{x^3y}{z}\right)^{1/2} = \frac{1}{2}\ln\frac{x^3y}{z} =$

$= \frac{1}{2}(\ln x^3y - \ln z) = \frac{1}{2}(\ln x^3 + \ln y - \ln z) =$

$= \frac{1}{2}(3\ln x + \ln y - \ln z) = \boxed{\frac{3}{2}\ln x + \frac{1}{2}\ln y - \frac{1}{2}\ln z}$

ex. Rewrite and simplify:

1) $\ln \dfrac{\sqrt{x^3 e^{x-1}}}{x^2+1} = \ln\left(x^3 e^{x-1}\right)^{1/2} - \ln(x^2+1) =$

$= \frac{1}{2}\left(\ln x^3 e^{x-1}\right) - \ln(x^2+1) =$

$= \frac{1}{2}\left(\ln x^3 + \ln e^{x-1}\right) - \ln(x^2+1) =$

$= \frac{1}{2}\left(3\ln x + (x-1)\underbrace{\ln e}_{=1}\right) - \ln(x^2+1) =$

$= \frac{1}{2}(3\ln x + x - 1) - \ln(x^2+1) = \boxed{\frac{3}{2}\ln x + \frac{1}{2}x - \frac{1}{2} - \ln(x^2+1)}$

2) $\log \sqrt{x\sqrt{y\sqrt{z}}} =$

$= \log\left(x\sqrt{y\sqrt{z}}\right)^{1/2}$

$= \frac{1}{2}\log\left(x\sqrt{y\sqrt{z}}\right)$

$= \frac{1}{2}\left(\log x + \log\sqrt{y\sqrt{z}}\right)$

$= \frac{1}{2}\left(\log x + \log\left(y\sqrt{z}\right)^{1/2}\right)$

$= \frac{1}{2}\left(\log x + \frac{1}{2}\log y\sqrt{z}\right)$

$= \frac{1}{2}\left(\log x + \frac{1}{2}\left(\log y + \log\sqrt{z}\right)\right)$

$= \frac{1}{2}\left(\log x + \frac{1}{2}\left(\log y + \frac{1}{2}\log z\right)\right)$

$= \frac{1}{2}\left(\log x + \frac{1}{2}\log y + \frac{1}{4}\log z\right)$

$= \boxed{\frac{1}{2}\log x + \frac{1}{4}\log y + \frac{1}{8}\log z}$

ex. Write as a single logarithm:

1) $\frac{1}{2}[\ln(x-5)+\ln x]-\ln(2y) =$

$\frac{1}{2}\left[\ln(x-5)(x)\right]-\ln 2y =$

$= \ln\sqrt{(x-5)(x)} - \ln 2y =$

$$\boxed{\ln \frac{\sqrt{(x-5)(x)}}{2y}}$$

2) $\log 2x - \log(x+1) - \frac{1}{3}\log(3x+7)$

$= \log \frac{2x}{x+1} - \log(3x+7)^{1/3} =$

$= \log \frac{\frac{2x}{x+1}}{\sqrt[3]{3x+7}} = \boxed{\log \frac{2x}{(x+1)\sqrt[3]{3x+7}}}$

$\frac{1}{2} = \frac{2}{4}$ $\quad \log_2 5 = \frac{\log 5}{\log 2} = \frac{\ln 5}{\ln 2}$

Change of Base Formula

Let $a \neq 1$, $b \neq 1$ and x be positive real numbers. Then

Choose any base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

When using a calculator, we need the specific formulas:

$$\log_a x = \frac{\log x}{\log a} \quad \text{or} \quad \log_a x = \frac{\ln x}{\ln a}$$

base 10 $\qquad$ base e

NOTE:

Most calculators have both 'log' and 'ln' keys to calculate the common and natural logarithm of a number. By using the Change of Base Formula, we can

1. evaluate logarithms to other bases.
2. graph logarithms to other bases.

ex. Given $\log 5 \approx 0.7, \log 3 \approx 0.48$, $\ln 3 \approx 1.1$, and $\ln(4+e) = 1.9$, use Change of Base Formula to find:

1) $\log_3 5 = \dfrac{\log 5}{\log 3} = \boxed{\dfrac{0.7}{0.48}}$

2) $\log_{\sqrt{3}} \sqrt{4+e} = \dfrac{\ln \sqrt{4+e}}{\ln \sqrt{3}} = \dfrac{\cancel{\frac{1}{2}} \ln(4+e)}{\cancel{\frac{1}{2}} \ln 3} = \boxed{\dfrac{1.9}{1.1}}$

ex. Solve $\log_3 x = \log_9(2x-1)$ change to base 3

$\log_3 x = \dfrac{\log_3(2x-1)}{\log_3 9}$ $= 2$

$\log_3 x = \dfrac{\log_3(2x-1)}{2}$

$2\log_3 x = \log_3(2x-1)$

$\log_3 x^2 = \log_3(2x-1)$

$x^2 = 2x-1$

$x^2 - 2x + 1 = 0$

$(x-1)(x-1) = 0$

$x = 1$ and it is in the domain

Practice.

1) Write $\log_4 x + 4\log_2 y$ as a single logarithm with base 2.

2) Solve: $2\log_3 x = \log_9 16$

3) Solve: $\log_2 x = \log_4 25$

4) Solve: $\log_5 x = \log_{\sqrt{5}} 6$

Answer. 1) $\log_2 \sqrt{x}y^4$ 2) $x = 2$ 3) $x = 5$ 4) $x = 36$